

POAMS: LECTURE 12
THE DISCRETE NATURE OF LIGHT- ENERGY FROM
THE NORMAL REALIST PERSPECTIVE

In this lecture, I will continue to present some of our ideas on the nature of light, particularly at the quantum level. After a recap on the discrete nature of light-energy, I will move on to discuss causality at the quantum level, photum interaction and conservation of four-momentum. This will be followed by our Normal Realist derivation of the Balmer-Rydberg formula. This lecture is again based on some of 'Light Speed...' Chapter 5.

1. The discrete nature of light-energy

As discussed in previous lectures, according to our Normal Realist philosophy, *light* is to be thought of as *observational information*, rather than consisting of 'particles' or 'waves', 'travelling at speed c '. Stripped of all its mechanistic precepts, light quanta are simply informational bits, in bar-code-like arrangements, out of which, ultimately, our knowledge of the world is constructed.

Further, according to the Planck relation, electromagnetic radiation with frequency ν is emitted and absorbed in whole numbers of quanta of energy, E , where

$$E = h\nu \quad (1)$$

and h is Planck's constant. We use the word *photum* (plural *phota*) for a quantum of transferable energy from one observational location to another since this carries no mechanical overtones of the usual ballistic kind. Recall that apart from its lack of the usual separatist, space-travelling connotations, the photum is the same in all other respects as the Einstein *photon*.

In our Normal Realist approach, the photum is treated as a single irreducible quantum event common to both an 'object' and the observer. There is no proper time separating whatever we may think of as the 'beginning and end' of a quantum interaction. Space and time, as we normally perceive them, therefore, cannot exist on that quantum level. These dimensions of physical phenomena exist solely on the macro-

phenomenal level, optically projected from informational patterns and sequences of these quantum ‘pixels’. On the quantum-phenomenal level, the atoms of bodies interact by direct resonance, with nothing existing or happening in between. [*End of recap*]

2. Causality at the quantum level

We maintain that there is no such thing as cause and effect at the quantum level of phenomenal analysis. A quantum event does not *cause* another, any more than one letter in this sentence causes the next in the sequence. So, what does ‘causality’ mean in this context? On its most mechanical level, cause and effect is the transfer of energy from a body in a higher energy-state to another body in a lower energy-state, according to the Second Law of Thermodynamics. In the classical thermodynamics of Ludwig Boltzmann (1844-1906) and others, these transfers are conducted by material particles in the void or by waves in some kind of medium. Since the advent of Information Theory, however, that Second Law has been redefined in terms of pure probability-statistics which apply not only to mechanical processes such as the transfer of heat from one body to another, but to all forms of *communication* between humans and similar organisms. Like distance and time, therefore, cause and effect are a function of statistical numbers of quantum events occurring in vast and fine-grained profusion. Since this includes communication with nature in general on all levels known to science, whether classified as organic or inorganic, it breaks down the traditional Cartesian dichotomy between ‘Mind’ and ‘Matter’.

None of this excludes informational exchanges between things other than ourselves; exchanges that may take place beyond our reach and experience. This is by no means makes the existence of these things ‘mysterious’ to us in any deep metaphysical sense. Their unseen and unknown character is a logically necessary extension of all that lies right there in front of us.

3. Photum interaction and conservation of four-momentum

In lecture 9, I discussed some relativistic dynamics, in particular the *four-momentum* of a material particle, the *relativistic mass* of such a particle, *conservation of four-momentum*, and the equivalence of mass and energy.

To recap some definitions, let P be a material particle of *constant rest mass*, m_0 . Let P have velocity $\mathbf{v}(t)$ and speed $v(t) = \|\mathbf{v}(t)\|$ relative to some observer O . Relative to the observer O , the *relativistic mass*, $m(t)$, of P is defined by

$$m(t) = m_0 \gamma(t) = m_0 (1 - v^2(t)/c^2)^{-1/2} \quad (2)$$

[Lecture 9, equation (7)]. The *four-momentum*, $\mathbf{P}$, of P , a function of *proper time*, τ , is then given by

$$\mathbf{P}(\tau) = m(t)(1, \mathbf{v}(t)) \quad (3)$$

[Lecture 9, equation (8)].

Recall that the *total energy*, $E(t)$, of P relative to O is given by

$$E(t) = m(t)c^2 \quad (4)$$

[Lecture 9, equation (19)] It follows by this equivalence of mass and energy and equation (3) that

$$\mathbf{P}(\tau) = \frac{E(t)}{c^2} (1, \mathbf{v}(t)) \quad (5)$$

From this result it follows that since any photum is essentially a quantum of energy, we may define the **four-momentum** of a photum with energy $E(t)$ relative to O , by

$$\mathbf{P} = \frac{E}{c^2} (1, c\mathbf{n}) \quad (6)$$

where $\mathbf{n}$ is a vector of unit length in the direction of the emitting source. Using equation (6), problems in which material particles interact via photum can be solved by applying conservation of four-momentum. Once again,

however, it is necessary to emphasise here that in our approach, a photum does not travel in the usual sense but is instantaneously emitted and absorbed in its own proper time as a discrete packet of energy E .

Example

Suppose that an elementary particle radiates electromagnetic energy in the form of a photum whilst at rest relative to some observer and that, as a result, the particle's rest mass is reduced by one fifth. We may calculate the particle's speed of recoil and hence the energy of the photum, in terms of the particle's rest mass, by applying conservation of four-momentum as follows.

Let m_0 denote the rest mass of the elementary particle and let the energy of the photum be E . Choose Cartesian coordinates so that the elementary particle travels in the x -direction after emission, relative to the observer. It then follows by equations (2), (3) and (6) together with conservation of four-momentum that,

$$m_0(1, 0, 0, 0) = \frac{E}{c^2}(1, c, 0, 0) + \frac{4}{5}\gamma(v)m_0(1, v, 0, 0)$$

where $v(t)$ is the speed of the resulting particle. Hence, by comparing components in this equation,

$$m_0c^2 = E + \frac{4}{5}\gamma(v)m_0c^2 \quad (7) \quad \text{and} \quad 0 = \frac{E}{c} + \frac{4}{5}\gamma(v)m_0v \quad (8)$$

Equation (7) expresses conservation of energy and equation (8) expresses conservation of relativistic momentum. By eliminating E from (7) and (8) it follows that,

$$\begin{aligned} m_0c^2 &= \frac{4}{5}\gamma(v)m_0c^2 - \frac{4}{5}\gamma(v)m_0vc \Rightarrow 5c = 4\gamma(v)(c - v) \\ \Rightarrow 25c^2 &= \frac{16c^2(c - v)^2}{c^2 - v^2} \Rightarrow 25(c + v) = 16(c - v) \Rightarrow 41v = -9c \end{aligned}$$

The resulting particle's speed of *recoil* is therefore $9c/41$. Then by equation (8), the energy of the photum is

$$E = -\frac{4}{5}\gamma(v)m_0vc = -\frac{4}{5}(1 - 81/41^2)^{-1/2}\left(-\frac{9}{41}m_0c^2\right)$$

$$\Rightarrow E = \frac{4}{5} \cdot \frac{41}{40} \cdot \frac{9}{41} m_0c^2 = \frac{9}{50} m_0c^2$$

From this example it can be seen that it is a fallacy to take the fact of 'photonic' recoil and transfer of momentum as 'evidence' of the ballistic or 'photonic' nature of the light-quantum, since this fact can be fully explained without ascribing any real velocity to that quantum. This example also demonstrates that different photons may have different energies. Traditionally this fact is expressed by the Planck relation, equation (1).

4. A Normal Realist derivation of the Balmer-Rydberg formula

The Planck relation, equation (1), asserts that electromagnetic radiation consists of a whole number of quanta of energy $h\nu$. This suggests that there is a direct deductive connection between the energy released by an object in optical interaction and the discrete lines that are observed in atomic spectra, which are described by the (generalised) Balmer-Rydberg formula,

$$\nu(n_1, n_2) = c\alpha_N R N^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \quad (9)$$

first discussed in Lecture 10. [*Lecture 10, equation (3)*] In this formula, ν is the frequency of the light in cycles per second, N is the atomic number of the element (*e.g.*, $N = 1$ for hydrogen), R is a constant, known as the *Rydberg constant*, n_1 and n_2 are natural (whole) numbers representing the fixed and running terms respectively, and α_N is a constant depending on the element, with $\alpha_N \cong 1$. Bohr provided one such connection, as studied in section 3 of Lecture 10. However, this is not the only possible derivation of formula (9). In this section, I will provide a heuristic Normal Realist derivation of (9), which, in contrast to Bohr's, does not involve any

conception of ‘hidden mechanisms’ of electro-dynamics. Rather, our derivation is based on purely observational considerations.

It follows from the equivalence of mass and energy, first discovered by Einstein, discussed in Lecture 9, and expressed in equation (4), that if energy is discrete, then so is mass. This suggests that there is a natural, indivisible unit of mass in the phenomenon of electromagnetic radiation. Since, because of the Planck relation (1), frequency is also discrete, it seems reasonable to suppose that there is also a natural indivisible unit of time in this context. These measures are effectively continuous on the macro-scale, so there is no contradiction in supposing that the standard relativistic formulae are ultimately *discrete* on the micro-scale.

Consider again the standard *mass-increase-with-speed* formula,

$$m(t) = m_0 \gamma(t) = m_0 (1 - v^2(t)/c^2)^{-1/2} \quad (2)$$

Recall that here, m_0 is the rest mass of a material particle, $v(t)$ is its speed relative to an observer O , and $m(t)$ is its relativistic mass, as recorded by O . Suppose that, in accordance with our previous reasoning, there is some indivisible natural unit of mass, m^* say, for the phenomenon in question. Then at the quantum level, $m_0 = Nm^*$ and, at any particular time, $m = nm^*$ say, where N and n are natural (whole) numbers. Equation (2) then reads,

$$nm^* = (1 - v^2/c^2)^{-1/2} Nm^* \Rightarrow v^2/c^2 = 1 - N^2/n^2 \quad (10)$$

Thus, at the quantum level, $(v/c)^2$ takes discrete values only. It should be noted, however, that at this level, $(v/c)^2$ is no more than a terminal abstraction of the process of reduction, so it makes no sense to think of v , in this context, as a speed in the ordinary sense, any more than it makes sense to think of m^* as an ordinary mass. The ultimately discrete values of $(v/c)^2$ may also be obtained directly from the time-dilation formula [Lecture 5, equation (2)], by supposing that there is some indivisible unit of time.

For heuristic reasons, to be justified in due course, to each discrete value of v^2 may be assigned a total energy $\alpha_N m^* v^2$ (compare this with Einstein’s

equation (4) for the equivalence of mass and energy). Then equation (10) indicates that allowable discrete losses of energy in this context are given by

$$E(n_1, n_2) = \alpha_N m^* c^2 N^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \quad (11)$$

This formula implies that the traditional ‘atom’ is an accumulation of relativistic mass-energy in discrete amounts. In this sense, the atom may be thought of as a *quantum accumulator*. It follows by Planck’s relation (1), that $m^* c^2/h$ is some natural limit of frequency, which empirically, is cR , where R the Rydberg constant. It then follows that

$$m^* = \frac{hR}{c} \quad (12)$$

so that by equations (11) and (12), the allowable losses of energy are given by

$$E(n_1, n_2) = \alpha_N h c R N^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \quad (13)$$

The Balmer-Rydberg formula, equation (9), then follows directly from Planck’s relation (1).

Using the currently accepted values of $h \cong 6.626070 \times 10^{-34}$ J s, $R \cong 1.0973732 \times 10^7$ m⁻¹ and $c \cong 2.99792458 \times 10^8$ m s⁻¹, equation (12) then gives $m^* \cong 2.4254351 \times 10^{-35}$ kg. (Here J is the SI unit of a *Joule*, the kinetic energy of a 1 kg mass moving at 1 metre per second, so that J = kg m² s⁻².) We may associate this value of m^* with the ‘mass-equivalent’ of the energy of a *photum* transition associated with the hydrogen atom, since Bohr supposed that energy losses are radiated as single photons. We repeat, however, that in this derivation, the *photum*, unlike the photon, is not a space-travelling entity but a purely observational quantum of energy.

It follows from this discussion and equation (12) that the energy, E , of a photum transition from one atom to another is given by

$$E = m^* c^2 = h c R \cong 2.1798724 \times 10^{-18} \text{ J}$$

Conventionally, this is the energy equivalent of the ‘electron charge’, e , that is, $E = I_0 e$, where I_0 is the *ionisation potential*, the kinetic energy required to completely remove the electron from the atom. It must be borne in mind however, that in our approach, the electron is not conceived as a well-defined ‘electrically charged particle’.

Classically, a loss of energy must be produced by a movement of rest mass, m_0 , so that according to equation (17) of Lecture 9 and the discussion which follows that equation, in this context,

$$E = (m - m_0)c^2 = m^*c^2 \quad (14)$$

where m^* our fundamental unit of mass and m is the relativistic mass of the ‘electron’. In equation (14), the loss of energy is traditionally represented as a gain in kinetic energy of the electron. It then follows by equation (2) that

$$(1 - 1/\gamma)m = \left(1 - (1 - v^2/c^2)^{1/2}\right)m = m^* \quad (15)$$

where v is the conventional ‘speed’ of the electron. This equation can be used to derive the speed of the electron, given its relativistic mass. The currently accepted value of the relativistic mass of electron in the hydrogen atom is $m \cong 9.10938291 \times 10^{-31}$ kg. Substituting this value in equation (15), together with our calculated value of m^* gives $v \cong 2.187677 \times 10^6$ m s⁻¹. This is the currently accepted value of the ‘speed’ of the electron in the hydrogen atom in its ‘ground state’. With these approximate values, it follows that, using equation (12),

$$\frac{mv^2}{2} \cong m^*c^2 = hcR = E$$

In this sense, the energy loss can be approximately equated with the classically conceived kinetic energy of the moving electron, independent of any path it may be assumed to be following.

These derivations necessarily contain some *ad hoc* assumptions. It is nevertheless significant that some of the parameters of Bohr's historical model of the atom, as well as the Balmer-Rydberg formula, can be reproduced in this way without any references to 'electric charges' and associated 'field' and 'force' postulates. In later lectures, we continue our investigation into the parameters of the hydrogen atom, again avoiding any reference to 'charges', by supposing that angular momentum in all its forms is always conserved and ultimately quantised.

Beginning with the next lecture, I will now turn my attention to the phenomenon of true physical motion, which, according to Newton, is everywhere under the influence of the 'universal force of gravity', and hence never truly rectilinear, as studied in Special Relativity. Taking account of this essentially curvilinear character of motion extends our Normal Realist agenda to include what is normally known as General Relativity.

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